Real-Time Video Generation with MiniMax H3

8× RTX PRO 5000 · vLLM-Omni + ComfyUI

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Agenda

01 Toward Real-Time H3 Generation
02 vLLM-Omni + ComfyUI Applications

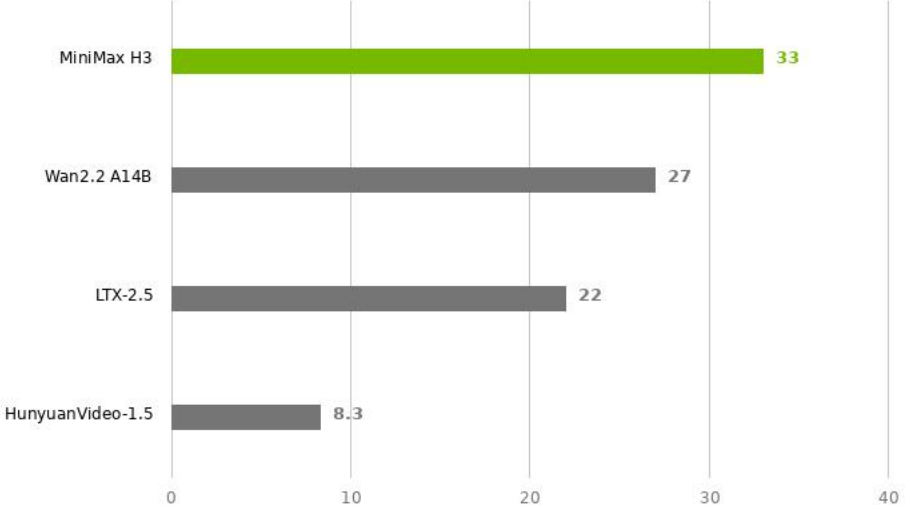


01 · Real-Time H3 Generation



Real-Time 768p Raises the Systems Bar

For 362 frames at 24 FPS, real time requires end-to-end latency < 15.1 s (RTF < 1.0)



GENERATIVE BACKBONE · FULL PARAMETERS (B)

RTF = latency / media duration. The real-time deadline here is 15.1 s. H3 is the largest dense backbone in this peer set; Wan2.2 = 27B total / 14B active.



OPEN H3-BASE · NATIVE 768P · 33B DENSE

32B encoder → 33B dense Omni Transformer → Visual/Audio VAE. 50 blocks × 50 denoising steps; full attention. ≈130 GB raw BF16; ~13B AdaLN cacheable.

Sources: MiniMax H3, Wan2.2, LTX-2.5 and HunyuanVideo-1.5 official releases · author 362-frame workload

Frontier Models · The Cards

Every discrete SM120 card combines fifth-generation Tensor Cores with FP4 and GDDR7; memory capacity and PCIe-to-fabric bandwidth define the deployment envelope

SKU	Compute	Memory	GPU I/O / Fabric
RTX PRO 5000 Blackwell	SM120 · 110 SMs 5th-gen Tensor · FP4	72 GB GDDR7 ECC 48 GB variant	PCIe Gen5 ×16 Planned: NIC → RDMA*
RTX PRO 6000 WS / Server	SM120 · 188 SMs 5th-gen Tensor · FP4	96 GB GDDR7 ECC	PCIe Gen5 ×16 No GPU NVLink
GeForce RTX 5090	SM120 · 170 SMs 5th-gen Tensor · FP4	32 GB GDDR7	PCIe Gen5 ×16 No NVLink
DGX Spark · GB10	SM121 · 48 SMs 5th-gen Tensor · FP4	128 GB unified LPDDR5X	NVLink-C2C in SoC 200 Gb/s ConnectX-7

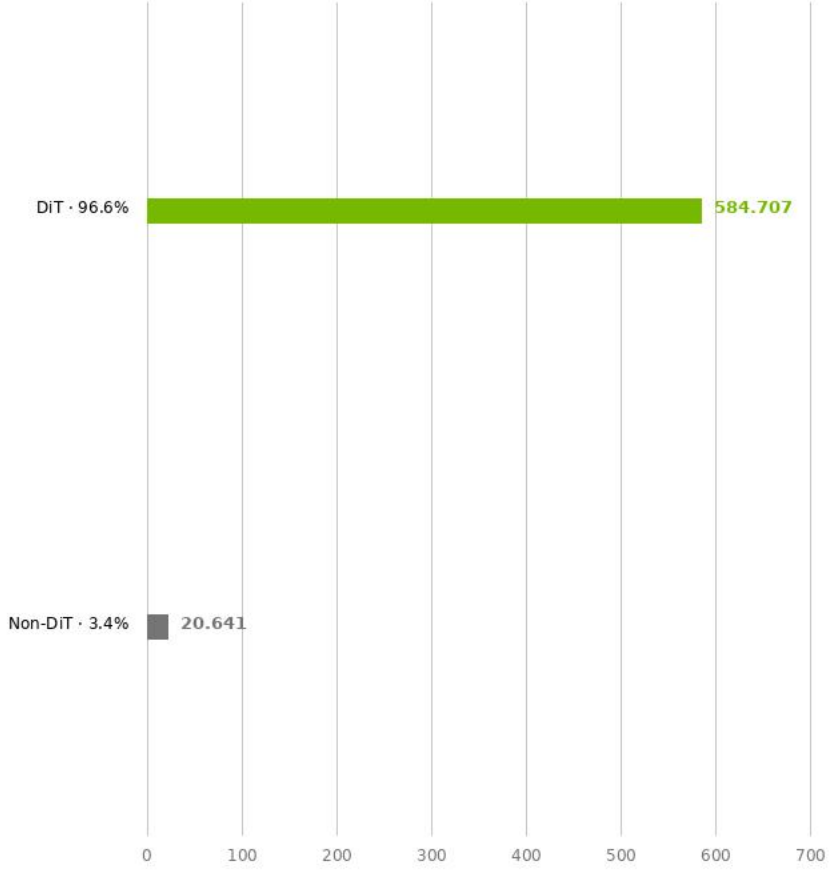
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* Planned system path; GPUDirect RDMA requires a compatible NIC and PCIe topology · Sources: NVIDIA product specs, CUDA GPU table, GPUDirect RDMA guide

Baseline Breakdown

8× RTX PRO 5000 · Original 50-step H3

Measured request · 605.348 s · RTF 40.1
DiT · 584.707 s · 96.6%
Non-DiT · 20.641 s · 3.4%
50 σ points → 49 DiT calls
Priority · fewer calls + lower attention cost



REQUEST LATENCY · 1280×704 / 362F @ 24 FPS
BF16 · TP1/SP8 · AdaLN cached · NCCL Ulysses · request-stage timing

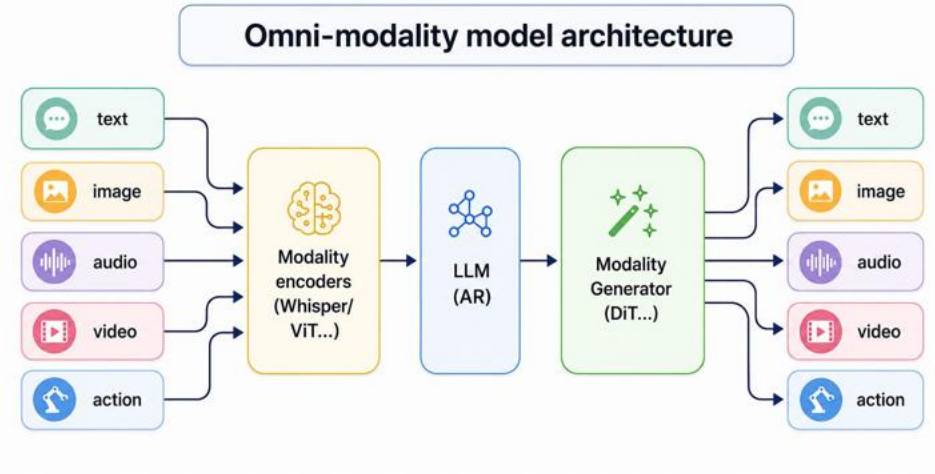
Author S1 request-stage timing · kernel/NCCL attribution requires Nsight

02 · vLLM-Omni ComfyUI Apps



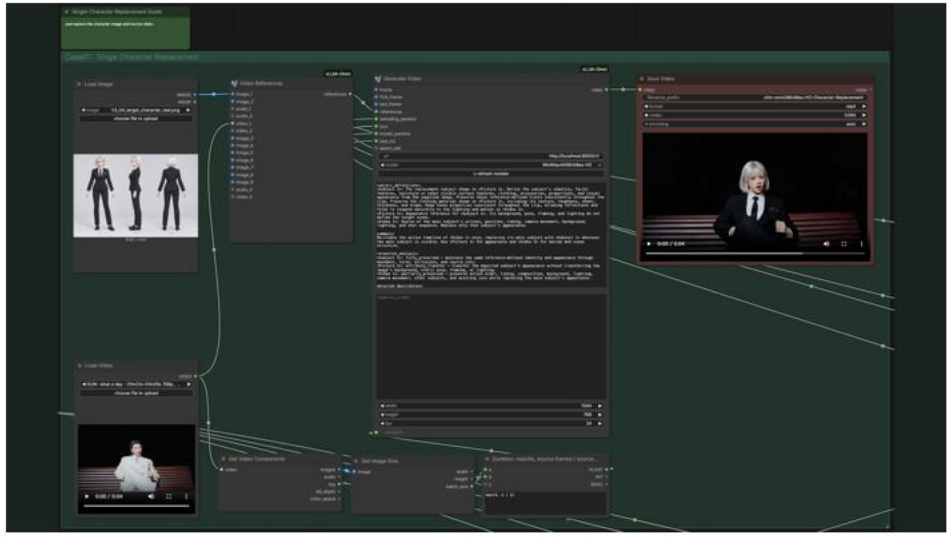
vLLM-Omni Connects ComfyUI to Multi-GPU H3 Serving

Open PR #9194: character replacement, continuation-aware references, and chainable LoRA controls



Official vLLM-Omni Architecture

ComfyUI remains the visual frontend; vLLM-Omni owns model execution, scheduling, and multi-GPU deployment



ComfyUI Workflow · PR #9194

Single- and multi-character replacement, source-derived duration, long-video continuation, and LoRA controls

Sources: vLLM-Omni v0.26 public deck, slide 22 · vLLM-Omni PR #9194 (open)